

STAT210B Lecture 1 The Non-Asymptotic Approach

Logic ▾

在学习具体内容之前, 我们先研究为什么要学习 non-asymptotic statistics, 以及 asymptotic statistics 有什么局限性

Quote ▾

为了说明学习 non-asymptotic statistics 的必要性, 引用 non-asymptotic statistics 先驱之一 Pascal Massart 的一段话:

"Through our works, we have promoted a nonasymptotic approach in statistics which consists in taking the number of observations as it is and trying to evaluate the effect of all the influential parameters. At this very starting point, it seems to me that it is important to provide a first answer to the following question: why should we be interested by a nonasymptotic view for model selection at all? In my opinion, the motivation should neither be a strange interest for 'small' sets of data nor a special taste for constants and inequalities rather than for limit theorems (although since mathematics is also a matter of taste, it is a possible way for getting involved in it...). On the contrary, the non-asymptotic point of view may turn to be especially relevant when the number of observations is large. It is indeed to fit large complex sets of data that one needs to deal with possibly huge collections of models at different scales. The non-asymptotic approach for model selection precisely allows the collection of models together with their dimensions to vary freely, letting the dimensions be possibly of the same order of magnitude as the number of observations."

“通过我们的研究, 我们推动了一种统计中的非渐近方法, 其核心思想是: 把观测数量当作给定的事实, 并尝试评估所有重要参数的影响。在这一出发点上, 我认为有必要首先回答这样一个问题: 为什么在模型选择中需要 non-asymptotic 的视角?”

在我看来, 动机既不是因为我们对‘小数据集’有特殊兴趣, 也不是因为我们偏好常数或不等式而非极限定理 (尽管数学品味确实因人而异)。

相反, non-asymptotic 的视角在观测数量很大时反而尤为重要。因为正是在处理大型、复杂的数据集时, 我们需要面对在不同尺度上可能极其庞大的模型集合。非渐近模型选择方法, 正是允许模型集合及其维度自由变化, 甚至使维度与观测数量处于同一数量级。”

1 学习 Non-Asymptotic Statistics 的动机

1.1 一个 asymptotic 的例子: Confidence Interval

令:

- $X_1, \dots, X_n$ 为 i.i.d r.v.
- $\mathbb{E}[X] = \mu$
- $\text{Var}[X_1] = \sigma^2$

则根据 central limit theorem, mean μ 有以下 confidence interval:

$$\mu \in \left[\frac{\sum_{i=1}^n X_i}{n} - \frac{2\sigma}{\sqrt{n}}, \frac{\sum_{i=1}^n X_i}{n} + \frac{2\sigma}{\sqrt{n}} \right]$$

其有 $1 - 2\Phi(-2) \approx 0.9545$ 的概率覆盖 true μ

且 μ 满足:

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma\sqrt{n}} \right| \geq t \right) = \mathbb{P}(|Z| \geq t) = 2\Phi(-t)$$

但使用 CLT 会存在以下问题:

- 仅在 $n \rightarrow \infty$ 时近似, 且难以判断 n 要多大才合理
- 仅允许 sample size n 变大, 要求其他参数, 如 σ , 在 n 变化时保持不变

⚠ Remark ▾

相较于 CLT, **non-asymptotic** 给的是 "有限样本误差控制", 形如

$$\mathbb{P}\left(|\bar{X} - \mu| \geq (\text{显式函数: } \sigma, n, \delta, \text{尾部条件})\right) \leq \delta$$

并且**明确依赖于分布尾部/矩条件** (如 sub-Gaussian, finite variance but heavy tail, bounded, sub-exponential 等), 从而可以计算 "这个分布下要达到 95% 覆盖大概需要什么量级的 n "

1.2 一个 non-asymptotic 的例子: Linear Regression

令:

- $Y_i = x_i^\top \theta^* + \xi_i, i \in \{1, \dots, n\}$ 且 i.i.d
- θ^* 为 target parameter
- $\xi_i \sim \mathcal{N}(0, \sigma^2)$

则 least square estimator 为:

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^n} \frac{1}{n} \sum_{i=1}^n (Y_i - x_i^\top \theta)^2$$

我们可以证明一个 non-asymptotic 的 concentration result:

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (x_i^\top \theta - x_i^\top \theta^*)^2 \right] \leq \frac{\sigma^2 d}{n}$$

其中参数可以随 n 的变化而变化, 例如 $d = d(n)$, 这是 asymptotic statistics 不允许的

我们可以更进一步, 推出形式如下 confidence set:

$$\left\{ \theta \in \mathbb{R}^d : \frac{1}{n} \sum_{i=1}^n (x_i^\top \hat{\theta} - x_i^\top \theta^*)^2 \leq \alpha_{n,d,\delta} \right\}$$

该 confidence set 对每一个给定的 n, d, δ 都成立, 不需要假设参数维度固定或 $n \rightarrow \infty$

≡ Example: 一个 d 会随 n 变化的例子 ▾

考虑医疗数据, 随着样本的累计, 我们意识到需要统计更多 features, 因此增加维度

2 Basic Tail Bounds: From Markov to Chernoff

Logic ▾

大多数 concentration bounds 都始于对 Markov's inequality 的 refined application

我们首先回顾两个经典不等式, 然后解释为什么 exponential moments 可以得到更强的界.

2.1 Proposition: Markov's Inequality

令:

- X 为 **non-negative** random variable
- $\mathbb{E}[X] < \infty$

则对任意 $t > 0$, 有:

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}$$

Proof ▾

由于 $X \geq 0$, 有

$$X \geq t \mathbf{1}\{X \geq t\}$$

取期望, 有

$$\begin{aligned} \mathbb{E}[X] &\geq t \mathbb{P}(X \geq t) \\ \Rightarrow \mathbb{P}(X \geq t) &\leq \frac{\mathbb{E}[X]}{t} \end{aligned}$$

2.2 Proposition: Chebyshev's Inequality

令:

- X 为 random variable
- $\mathbb{E}[X] < \infty$
- $\text{Var}[X] < \infty$

则对任意 $t > 0$, 有:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\text{Var}[X]}{t^2}$$

Proof ▾

为了便于表示, 令 $\mu = \mathbb{E}[X]$, 则

$$\begin{aligned} \mathbb{P}(|X - \mu| \geq t) &= \mathbb{P}((X - \mu)^2 \geq t^2) \\ &\leq \frac{\mathbb{E}[(X - \mu)^2]}{t^2} \quad (\text{Markov's inequality}) \\ &= \frac{\text{Var}[X]}{t^2} \end{aligned}$$

⚠ Remark: Chebyshev's Inequality 的 generalization ▾

令 $p > 1$ 为任意 constant, 则

$$\begin{aligned}\mathbb{P}(|X - \mu| \geq t) &= \mathbb{P}(|X - \mu|^p \geq t^p) \\ &\leq \frac{\mathbb{E}[|X - \mu|^p]}{t^p}\end{aligned}$$

因此 Chebyshev's inequality 可以被 generalize 为:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \inf_{p \geq 1} \frac{\mathbb{E}[|X - \mu|^p]}{t^p}$$

Logic ▾

对于 light-tail distribution 如 Gaussian, Chebyshev's inequality 并不足够 tight

2.3 Proposition: Tails of Gaussian Random Variables

令 $Z \sim \mathcal{N}(0, 1)$,

则对任意 $t > 0$, 有:

$$\left(\frac{1}{t} - \frac{1}{t^3}\right) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) \leq \mathbb{P}(Z > t) \leq \frac{1}{t} \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right)$$

⚠ Remark: 为什么需要这个 inequality? ▾

尽管 Gaussian tail 可以写作 $\mathbb{P}(Z \geq t) = \int_t^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx$, 但积分形式较难计算, 现实中使用 upper bound 往往已经够用

⚠ Remark: 和 Chebyshev's inequality 相比有什么优势? ▾

收敛速度为指数级, 比 Chebyshev's inequality 的 quadratic convergence 更快

🔗 Proof: upper bound 的证明 ▾

$$\begin{aligned}\mathbb{P}(Z \geq t) &= \int_t^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx \\ &= \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2} - yt - \frac{t^2}{2}\right) dy \quad (\text{令 } y = x - t, \text{ 即 } dx = dy, x = y + t) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) \int_0^\infty \exp\left(-yt - \frac{y^2}{2}\right) dy \\ &\leq \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) \int_0^\infty \exp(-yt) dy \\ &= \frac{1}{t} \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right)\end{aligned}$$

🔗 Proof: Lower bound 的证明 ▾

定义函数

$$f(x) = \left(\frac{1}{x} - \frac{1}{x^3}\right) e^{-x^2/2}.$$

直接计算可得

$$f'(x) = -(1 - 3x^{-4})e^{-x^2/2}.$$

因此

$$\int_t^\infty (1 - 3x^{-4})e^{-x^2/2} dx = f(t).$$

由于

$$1 - 3x^{-4} \leq 1,$$

得到

$$\int_t^\infty e^{-x^2/2} dx \geq \left(\frac{1}{t} - \frac{1}{t^3} \right) e^{-t^2/2}.$$

乘以 $1/\sqrt{2\pi}$ 得到下界.

Logic ▾

对于 Gaussian, 我们可以直接计算尾部, 而对于一般随机变量, 对应的技巧是使用 **exponential moments**, 由此引出 Chernoff's method

2.4 Definition: Moment Generating Function

令:

- X 为 random variable,
 - $\lambda \in \mathbb{R}$
- 则 moment generating function (MGF) $M_X(\lambda)$ 被定义为 real-valued function

$$M_X(\lambda) = \mathbb{E}[e^{\lambda X}]$$

其性质包括:

- 若 $X_1, \dots, X_n$ 为 independent, 则 $M_{X_1 + \dots + X_n}(\lambda) = \prod_{i=1}^n M_{X_i}(\lambda)$
- 若积分微分可交换, 则

$$\frac{d}{d\lambda^k} M_X(\lambda) = \frac{d}{d\lambda^k} \mathbb{E}[\exp(\lambda X)] = \mathbb{E} \left[\frac{d}{d\lambda^k} \exp(\lambda X) \right] = \mathbb{E}[X^k \exp(\lambda X)]$$

令 $\lambda = 0$, 则

$$\left. \frac{d}{d\lambda^k} M_X(\lambda) \right|_{\lambda=0} = \mathbb{E}[X^k]$$

Remark ▾

一个常用的技巧是对 MGF 在 0 处做 Taylor expansion, 有:

$$M_X(\lambda) = \mathbb{E}[\exp(\lambda X)] = \mathbb{E} \left[1 + \lambda X + \frac{\lambda^2 X^2}{2} + \dots \right]$$

该技巧可以用于判断某个分布族的 MGF 是否存在

2.5 Proposition: Chernoff's method

令:

- X 为 random variable
- X 的 MGF 为 M_X
- $\lambda > 0$ 为 fixed value

则:

$$\begin{aligned}
\mathbb{P}(X \geq t) &= \mathbb{P}(\lambda X \geq \lambda t) \\
&= \mathbb{P}(\exp(\lambda X) \geq \exp(\lambda t)) \\
&\leq \frac{\mathbb{E}[\exp(\lambda X)]}{\exp(\lambda t)} \\
&= \frac{M_X(\lambda)}{\exp(\lambda t)}
\end{aligned}$$

⚠ **Remark: Gaussian tail bound 和 Chernoff's method 的比较** ✓

令 $Z \sim \mathcal{N}(0, 1)$, 则其 MGF 为 $M_Z(\lambda) = \exp\left(\frac{\lambda^2}{2}\right)$

若使用 Gaussian tail upper bound, 有

$$\mathbb{P}(Z > t) \leq \frac{1}{t} \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right)$$

若使用 Chernoff's method, 则有

$$\mathbb{P}(Z \geq t) \leq \inf_{\lambda > 0} \frac{M_Z(\lambda)}{\exp(\lambda t)} = \inf_{\lambda > 0} \exp\left(\frac{\lambda^2}{2} - \lambda t\right) = \exp\left(-\frac{t^2}{2}\right) \quad (\text{令 } \lambda = t)$$